

CONTINUITY & DIFFERENTIABILITY

- A function $f(x)$ is defined on neighborhood of 'a' and satisfies the following:

- $\lim_{x \rightarrow a^-} f(x)$ exist and $\lim_{x \rightarrow a^-} f(x) = f(a)$, then $f(x)$ is left continuous at $x = a$.
- $\lim_{x \rightarrow a^+} f(x)$ exist and $\lim_{x \rightarrow a^+} f(x) = f(a)$, then $f(x)$ is right continuous at $x = a$.
- If $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = f(a)$,

Then $f(x)$ is **continuous** at $x = a$.

- Geometrical Meaning of Continuity:**

- The function $f(x)$ will be continuous at $x = a$ if there is no break in the graph of the function $y = f(x)$ at the point $(a, f(a))$.
- The function $f(x)$ will be continuous in the closed interval $[a, b]$ if the graph of the function $y = f(x)$ is an unbroken line or curved line from $(a, f(a))$ to $(b, f(b))$.

- If $f(x)$ is not continuous at $x = a$, then $f(x)$ is called as discontinuous at $x = a$.
- If $f(x)$ is discontinuous at $x = a$, then either of the following is true.

- $\lim_{x \rightarrow a^-} f(x)$ not exist.
- $\lim_{x \rightarrow a^+} f(x)$ not exist.
- $\lim_{x \rightarrow a^+} f(x) \neq f(a)$
- $\lim_{x \rightarrow a^-} f(x) \neq f(a)$
- $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$.

- Continuity of a function in an open interval (a, b):** A function $f(x)$ is said to be continuous in (a, b) if $f(x)$ is continuous at each and every point belongs to (a, b).

- Continuity of a function in a closed interval [a, b]:** A function $f(x)$ is said to be continuous in [a, b] if:

- $f(x)$ is continuous in the open interval (a, b).
- $f(x)$ is continuous at $x = a$ from the right.
- $f(x)$ is continuous at $x = b$ from the left.

- Algebra of Continuous Functions:** Let $f(x)$ and $g(x)$ be any two functions having $x = a$ as one of the condensation point of domain..

Let us define $h_1(x) = c_1 f(x) \pm c_2 g(x)$, $h_2(x) = f(x)g(x)$, $h_3(x) = f(x)/g(x)$.

- (1) When $f(x)$ and $g(x)$ both are continuous at $x = a$, then $h_1(x)$, $h_2(x)$, $h_3(x)$ will be continuous at $x = a$.
 - (2) When $f(x)$ or $g(x)$ is discontinuous at $x = a$, then $h_1(x)$ will be discontinuous at $x = a$ and we cannot say about the continuity of $h_2(x)$, $h_3(x)$, they may be or may not be continuous.
 - (3) When $f(x)$ and $g(x)$ both are discontinuous at $x = a$, then we cannot say about the continuity of $h_1(x)$, $h_2(x)$, $h_3(x)$, they may be or may not be continuous.
- **Accumulation Point/ Cluster Point/ Limit Point/ Condensation Point:** Accumulation point of a set of points is a point P such that there is atleast one point of the set distinct from P in any nbd of the given point; a point which is the limit of a sequence of points of the set. An accumulation point of a sequence is a point P such that there are an infinite number of terms of the sequence in any nbd of P.
 - **Note Points:**
 - (1) Every Polynomial function is continuous on $\mathbb{R}$ (real number set).
 - (2) Every constant function is continuous on $\mathbb{R}$ (real number set).
 - (3) Every Identity function is continuous on $\mathbb{R}$ (real number set).
 - (4) $\sin x$ is continuous on $\mathbb{R}$ (real number set).
 - (5) $\cos x$ is continuous on $\mathbb{R}$ (real number set).
 - (6) e^x is continuous on $\mathbb{R}$ (real number set).
 - (7) $|x|$ is continuous on $\mathbb{R}$ (real number set).
 - (8) $\log x$ is continuous on $\mathbb{R}^+$ (Positive real number set).
 - (9) $\log|x|$ is continuous on $\mathbb{R} - \{0\}$ (real number set).
 - (10) $\tan x$ is continuous on $\mathbb{R} - \{(2n+1)\frac{\pi}{2}; n \in \mathbb{Z}\}$.
 - (11) $\sec x$ is continuous on $\mathbb{R} - \{(2n+1)\frac{\pi}{2}; n \in \mathbb{Z}\}$.
 - (12) $\cot x$ is continuous on $\mathbb{R} - \{n\pi; n \in \mathbb{Z}\}$.
 - (13) $\operatorname{cosec} x$ is continuous on $\mathbb{R} - \{n\pi; n \in \mathbb{Z}\}$.
 - (14) $[x]$ is continuous on $\mathbb{R} - \mathbb{Z}$.
 - (15) $x - [x]$ is continuous on $\mathbb{R} - \mathbb{Z}$.
 - (16) $f(x)$ is continuous at $x = a$, then $|f(x)|$ is continuous at $x = a$, but converse is not true.
 - (17) Every differentiable function is continuous but converse is not true. Example: $|x|$ is continuous at $x = 0$ but not differentiable at $x = 0$.
 - (18) If f and g are continuous at a point $x = a$, then $f + g$, $f - g$, fg , f/g are also continuous at $x = a$.
 - (19) If f and g are continuous at a point $x = a$, then $f \circ g$ is also continuous at $x = a$.
 - (20) Each inverse trigonometric function is continuous in its domain.

• **Removal Discontinuity:**

(1) $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$, limit exist.

(2) $\lim_{x \rightarrow a} f(x) \neq f(a)$

• **Types of Removal Discontinuity:**

(1) **Missing Point Discontinuity:** For a function $f(x)$, $\lim_{x \rightarrow a} f(x)$ exist finitely but $f(a)$ is not defined. Then $x = a$ is called as missing point discontinuity.

(2) **Isolated Point Discontinuity:** For a function $f(x)$, $\lim_{x \rightarrow a} f(x) \neq f(a)$, then $x = a$ is called as isolated point discontinuity.

• **Irremoval Discontinuity or Jump Discontinuity:**

(1) $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$, limit does not exist, $f(a)$ may exist or may not exist.

• **Types of Irremoval Discontinuity:** In case of Irremoval discontinuity $f(a)$ may exist or may not exist but $\lim_{x \rightarrow a} f(x)$ does not exist.

(1) **Finite Discontinuity:** Here $f(a)$ exist finitely, but it is either equal to $\lim_{x \rightarrow a^+} f(x)$ or $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$, $\lim_{x \rightarrow a^-} f(x)$ are finite and unequal.

(2) **Infinite Discontinuity:** Here both $f(a)$ and $\lim_{x \rightarrow a} f(x)$ does not exist.

(3) **Oscillatory Discontinuity:** For a function $f(x)$, if $f(a)$ is undefined as $f(x)$ oscillates finitely as $x \rightarrow a$, then $x = a$ is called as point of finite oscillatory discontinuous. If $f(a)$ is undefined as $f(x)$ oscillates infinitely as $x \rightarrow a$, then $x = a$ is called as point of infinite oscillatory discontinuous.

• **Properties of Continuous Function:** Let $y = f(x)$ be a continuous function $\forall x \in [a, b]$, then the following will hold true.

(1) $y = f(x)$ is bounded between a and b . It means we can find a real number m_1 and m_2 such that $m_1 \leq f(x) \leq m_2$, $\forall x \in [a, b]$.

(2) Every value between $f(a)$ and $f(b)$ will be assumed by the function atleast once.

(3) If $f(a)f(b) < 0$, then $f(x)$ will become zero atleast once in (a, b) . It means if $f(a)$ and $f(b)$ are of opposite sign then the equation $f(x) = 0$ will have atleast one root in (a, b) .
